PROCESS BOOK

VizLaLune

Milan-Cortina 2026 Winter Olympics

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PROJECT OVERVIEW

VizLaLune is an interactive web exploration of the **XXV Olympic Winter Games**, held in **Milano and Cortina** d'Ampezzo from the **6th to the 22nd of February 2026**.
Rather than rehearsing the familiar story of medal rankings, our site **investigates the structure of participation itself**: how many athletes compete, where they come from, how disciplines distribute them, and how gender representation has evolved to what is, by several measures, the most balanced Winter Olympics in history.
The final deliverable is a single-page scrollytelling website built with JS libraries such as D3, Leaflet, or TopoJSON, featuring a page-wide selection state shared across all visualizations. We targeted three reader profiles: *sports fans* curious about behind-the-scenes patterns, *data journalists* looking for narrative angles, and *readers interested in gender representation* in international competitions.

MOTIVATION

The dataset was compiled by Petro Ivaniuk on Kaggle, it includes **2,916 athletes from 93 countries** competing across **116 events in 16 disciplines**, hosted at **15 venues spread over 22,000 km²**.
Three properties made it particularly interesting:

Recency.
Milano-Cortina just concluded, and since the Winter Olympics only takes place every 4 years, it represents a meaningful state of the current fight for equal representation at the highest level.

A milestone for gender parity.
Milano-Cortina was the first Winter Olympics with 12 of 16 disciplines fully gender-equal and 47% female participation. Hence, it seems worth to explore what this singular value represents for the XXV Games.

Multiple natural angles.
Inequalities can span accross various angles. Geography, gender, disciplines can all be source to potential representation issues.

FROM SKETCHES ...

Our Milestone 1 sketches proposed a **three-part structure**: a global overview (Part 1), a geographic structure section (Part 2), and a gender & team representation section (Part 3), capped by an animated conclusion.
For Milestone 2, we expanded this into **9 figures** with formal justifications drawn from the lecture material (Munzner's nested model, Shneiderman's mantra, Segel & Heer on narrative visualization).

... TO FINAL DESIGN

Implementation forced us to **refine**, **simplify**, and in some cases **reinvent**.
The main difference between the M2 plan and the final website are:

Two sections instead of three.
The "Geographic" and "Gender" parts naturally merged: the world map already encodes *where* athletes came from, and toggling its colour by gender turned the same view into the entry point of the gender story. Keeping them apart felt artificial, merging them tightened the narrative flow.

Fig 1.3 (venue exploitation bar chart) removed.
Once the venue map was clickable and showed disciplines on hover, the parallel bar chart added little. In addition as schedules.csv was highly incomplete such graphic would not represent real venue usage.

Send-to-Milano animation.
M2 described "one dot per athlete travels from country of origin to host venues" as a passive animation. The final version exposes it as a user-triggered button, which feels more like a deliberate moment in the narrative and lets the user replay it.

Discipline grid added. Not in the M2.
We realised the discipline filter, while present as a dropdown in our prototype, lacked a visual entry point. A grid of the 16 official Milano-Cortina pictograms doubles as filter and as decoration. It grounds the page in the Games' own visual identity.

Fig 3.2 (individual vs. team) removed.
The diverging gender bars already carry a heavy informational load adding a second mirror chart would have crowded the section without telling a new story.

Medal-race animation.
What M2 called a "classic medal count" became a Feb 6 → Feb 22 race-bar animation with a replay/play buttons and slider.

Global filter badge.
M2 introduced the shared-state idea conceptually. The final UI is a fixed top-right filter badge with autocomplete on country and discipline with a reset button.

DESIGN DECISIONS

VISUAL IDENTITY

The palette : deep navy (#091F36), cyan (#1295AF) and warm off-white (#F2F1ED), is directly **inspired by the official Milano-Cortina 2026 visual identity**. Borrowing this palette anchors the site in the actual Games and saved us the slow and dangerous work of picking colours from scratch (see Week 6.1's warnings about colour misuse).
The serif display font (used in section titles and the banner) echoes the dignified register the Olympics tend to use in their own publications.

We deliberately avoided the temptation to colour-code countries or disciplines on every chart. Following Munzner's advice that colour is one of the most overused channels, we **reserve colour for cases where it carries semantic weight**:

- the discipline colours in the treemap (to anchor a hierarchy),
- the gold/silver/bronze in the medal race (a culturally fixed encoding), and
- the male-blue / female-pink pairing in the gender plots

USING THE OFFICIAL PICTOGRAMS

The Olympic pictograms are **public-facing artwork** commissioned by Milano-Cortina specifically to make disciplines instantly recognisable across languages. Reusing them was an obvious win: it gives the page a recognisable visual rhythm, lets us **replace text labels with images in several places** (the discipline grid, the venue map markers, the podium cards), and frees the user from having to decode "what does the Skeleton discipline look like?".

LAYOUT : SCROLLYTELLING ON A SINGLE PAGE

We chose a **single-page strorytelling layout** over a multipage or tabbed interface for two reasons. First, **our story is sequential** by design (overview → filter → details). Scrolling matches the reading metaphor. Second, a single-page layout lets the **global filter badges** it in a fixed corner and apply to every visualization without page reloads: the linked-views pattern stays alive throughout the experience.

A **sliding sidebar** provides **direct navigation** to each section for readers who want to jump around, and the banner ends with a downward "Explore" arrow that gives a clear, low-friction entry point.

INTERACTIVITY : HOVER, FILTER, CLICK

One of the hardest design questions was deciding *which* action should do *what*. We converged on three roles:

Hover = preview.
Tooltips on countries, disciplines, bars, and venues. No state change. Reversible by moving the mouse away.

Global filter (top-right badge) = persistent selection.
Typing a country or discipline in the badge updates *every* visualisation on the page that knows about that dimension : the bubbles update their counts, the map highlights the country, the bar chart filters, and so on.

Click = local selection.
Clicking a country on the world map updates the adjacent bar chart. Clicking a discipline card opens the podium table. Local actions may also update the global filter when appropriate.

This three-tier model lets a curious user **explore freely without losing their place**.
We considered making clicks always global, but the experience became disorienting, the page would jump and reorganise on every interaction. Keeping clicks local as much as possible and reserving the explicit filter for "I want this to stick" worked much better in testing.

LINKED VIEWS VIA A SHARED STATE

Every section reads from a single small piece of state similar to below:

```
const state = {
  selectedCountry: null,
  selectedDiscipline: null
};
```

Any interaction that modifies state triggers a cascade of update functions registered by each visualization (updateBubbles, updateBarChart, highlightCountryOnMap, etc.). This is the **linked-views pattern** from Week 5.1, and once in place it dramatically simplified development: each visualization only needed to know how to draw itself *given a state*, not how to talk to all its neighbours.

CHOOSING THE RIGHT IDIOM

Each figure went through a small idiom shortlist before being implemented. Some highlights:

Diverging bars for gender.
Pinned to a central axis, they make symmetry instantly readable. Any asymmetry literally pops out as a longer side.

Worldmap: choropleth + dots.
A pure choropleth would hide individual athletes, a pure dot map would lose the country totals. The hybrid encodes count via colour (per country) *and* individuals via dots (one per athlete), and the two views support each other.

Treemap (disciplines → events) was preferred over a sunburst or nested pie because rectangular area is the most accurate channel for quantitative comparison (Munzner Week 6.2).

IMPLEMENTATION CHALLENGES

BALANCING BEAUTY WITH INFORMATION

Accross the project, we kept asking ourselves: *should this view be **prettier or denser**?* The treemap could have shown every event label in a tiny font, we chose to truncate aggressively and reveal full names on hover. The world map could have shown one dot per athlete with full opacity, we chose translucent dots so the underlying choropleth stays readable. In almost every case we erred on the side of restraint, on the principle that **an unread label communicates nothing**.

SYNCHRONISING FILTERS ACROSS VISUALIZATIONS

With multiple visualizations and a global filter badge, every interaction could cascade to tens possible updates. We solved this by **centralising state in skeleton.js**, exposing each visualization's update function on window (e.g. window.highlightCountryOnMap), and making all update functions idempotent.

CHOOSING THE RIGHT INTERACTION MODEL

Related but distinct from the synchronisation challenge: deciding *which gestures meant what* took several iterations. We discovered, for example, that double-click on a discipline label in the venue popup felt natural for "deep dive", whereas single-click should update the global filter. We documented the convention in short helper texts ("*Single-click a venue to discover...double-click to learn more*") rather than relying on users to discover it.

D3.JS FROM SCRATCH

Two team members were complete D3 beginners at the start. The library's update pattern, functional style, and the conceptual gap between SVG and the DOM all required a real ramp-up. Each visualization acted as a teaching example to the rest of the team.

DATA PREPROCESSING

The Kaggle dataset is rich but messy: country codes follow the Olympic three-letter convention (IOC, e.g. GER, NOR, AIN for the Individual Neutral Athletes), which doesn't align with the ISO codes used by the TopoJSON world atlas. We built a **manual mapping** (isoAlpha3ToNumeric) covering all 93 participating countries plus a few aliases. The events column was a stringified Python dict, which we parsed with a small regex rather than full JSON.parse. Coordinates for the dot map came from a hand-curated countries_coords.json.

FINAL VISUALIZATIONS

SECTION 1 - GLOBAL OVERVIEW

The first section answers ***what does this Olympics look like?*** then lets the reader zoom into any discipline or venue.

- **Intro bubbles**: Five animated circles displaying total counts of athletes, countries, disciplines, events, and venues - updated live when the global filter is applied.
- **Discipline grid**: The 16 official pictograms, each clickable to reveal a podium view below and set the global filter.
- **Podium & events table**: A sortable table of every event in the selected discipline: name, type, venue, date, and gold/silver/bronze winners.
- **Venue map**: A Leaflet map of northern Italy with the 15 venues as pictogram markers. Single-click to explore disciplines, double-click to open the official olympics.com page.
- **Treemap**: Disciplines → events, area proportional to √athletes. Cross-Country Skiing and Ice Hockey dominate, Ski Mountaineering sits small in the corner as the newest discipline.
- **Medal race**: A bar-chart animation racing through the 17 competition days, with a play/replay button as well as a slider.

SECTION 2 - GENDER REPRESENTATION

Section 2 follows the final state of women representation at Winter Olympics following a century-long progress

- **Opening narrative**: From 4.3% female athletes at Chamonix 1924 (Figure Skating only) to 47% across 12 fully gender-equal disciplines in 2026.
- **World map with athlete dots**: Countries coloured by athlete count, with one dot per athlete. A "*Send to Milano!*" button animates the global gathering. A gender toggle recolours dots blue/pink per country.
- **Linked bar chart:** Athletes per discipline, to the right of the map. Clicking a country filters it to that country's breakdown.
- **Symmetrical gender plot**: A diverging bar chart pinned to a central axis: men left in blue, women right in pink, with percentage splits per discipline. Fully linked to region filters. Nordic Combined's 100%-left bar is visually inescapable.

CONCLUSION

The site closes with two elements:

- **Conclusion text** : A short paragraph framing Milano-Cortina as a milestone on an ongoing journey, with a direct quotation from the Olympic Charter about the IOC's mandate to promote women in sport.
- **Conclusion bubbles** : An animated bubble field summarising the most striking insights gathered through the page. The bubbles enter the page with a soft physics-driven animation, leaving the reader with a final exploratory summary rather than a wall of text.

REFLECTIONS

WHAT WE ARE PROUD OF

The site holds together as a **single coherent piece**, rather than just charts on a page. The borrowed visual identity from Milano-Cortina makes it feel like an **extension of the Games** rather than a bland dashboard. The discipline grid + podium combination, in particular, surprised us by how much it sharpened the page: it turned a flat filter dropdown into something readers actively explored.

WHAT WE COULD HAVE DONE DIFFERENTLY

The most concrete mistake was **submitting the wrong prototype** for Milestone 2. This didn't reflect the actual state of our work at the time and cost us points we had been working on.
On the design side, we could have **started the cross-visualisation filter system earlier**. Connecting nine figures through a shared state turned out to be the hardest part of the project. We built each visualization in isolation first and added the links later, which led to refactoring and subtle bugs.

WHAT WE LEARNED

Working with D3.js from scratch taught us that the library rewards reading the docs carefully and punishes copy-paste from old StackOverflow answers. Working together across nine inter-linked visualizations taught us the **value of a tiny shared API** (one state object, idempotent update functions) over more elaborate abstractions.
And working under a deadline, with three sets of opinions, taught us that the fastest way through a design disagreement is usually to **mock both versions and look at them side by side.**

TEAM CONTRIBUTIONS

LÉONTINE LEFRANC

Mainly contributed to Section 2 (Gender Representation): the interactive world map showing each country's delegation as clusters of dots, and the horizontal bar chart of athletes per discipline linked to country selection. I also contributed scroll-triggered animations to the website, overall visualization structure, wrote the process book and the README, and participated equally in the screencast with Thomas and Manon.

MANON CESBRON DARNAUD

Contributed to the design reflection and ideation of the overall visualization structure, designed and implemented the treemap and the medal race visualizations, participated in the writing of the process book, handled the visual design of the process book, edited and assembled the screencast video

THOMAS LENGES

Explored multiple potential datasets to identify the most suitable sources for the project. Pre-processed and cleaned the data to ensure completeness and consistency. Designed the webpage skeleton, defining the overall structure including the sidebar, banner, footer, colour scheme, etc. Wrote the storytelling texts and structured the web page to guide users through the narrative in a coherent and engaging way.
Implemented global filters for discipline and country, enabling dynamic and personalised data exploration. Developed the intro bubbles, the discipline grid, and the associated discipline detail views. Built and integrated the venue map. Added discipline filtering to the treemap, and implemented the medal race play/pause button and the day selector slider.

Designed and implemented gender symmetric plots at both the general and discipline-specific levels. Produced the conclusion figure to close the narrative. Established the project's file structure and coded the mapping between country codes and their corresponding country names and continents. Finally, refined the world map and corresponding barchart filtering.

REFERENCE AND SOURCES

DATASET

Ivaniuk, P. *Milano-Cortina 2026 Olympic Winter Games.* Kaggle.

TOOLS & LIBRARIES

ID3.js v7 (d3js.org); Leaflet 1.9.4; TopoJSON 3.0.2; world-atlas (110m countries).

VISUAL IDENTITY

Palette inspired by the official Milano-Cortina 2026 brand guidelines. Discipline pictograms are the official Milano-Cortina 2026 pictograms, used here for non-commercial educational purposes.

COURSE LECTURE

W4.1Data; W4.2D3.js; W5.1Interaction; W5.2 More interactive D3; W6.1 Perception & colour; W6.2 Marks & channels; W7.2 Designing visualizations; W8.1–8.2 Maps; W10 Graph visualization; W11Tabular data; W12.1 Storytelling.

ACADEMIC REFERENCES

Munzner, T. (2014). *Visualization Analysis & Design.*
CRC Press. Shneiderman, B. (1996). The eyes have it: A task by data type taxonomy for information visualizations. *IEEE Symp. on Visual Languages.* Segel, E. & Heer, J. (2010). Narrative visualization: Telling stories with data. IEEE TVCG, 16(6). Johnson, B. *Proc.* IEEE *Visualization*.

ADDITIONAL CONTEXT

Britannica, Chamonix 1924 Olympic Winter Games. *Olympic Charter*, IOC, available at olympics.com.